

Exercise 2: Factory Method Pattern

Project: **FactoryMethodPatternExample**

Program Output (Console)

```
C:\FactoryMethodPatternExample> javac *.java  
C:\FactoryMethodPatternExample> java DocumentFactoryTest
```

```
Creating a Word Document...  
Opening Word Document (.docx)  
Word document created successfully.
```

```
Creating a PDF Document...  
Opening PDF Document (.pdf)  
PDF document created successfully.
```

```
Creating an Excel Document...  
Opening Excel Document (.xlsx)  
Excel document created successfully.
```

```
--- All documents created using Factory Method Pattern ---
```

Explanation

The Factory Method Pattern defines an interface (**DocumentFactory**) for creating an object but lets subclasses (**WordDocumentFactory**, **PdfDocumentFactory**, **ExcelDocumentFactory**) decide which class to instantiate. This decouples the client code (**DocumentFactoryTest**) from the concrete document classes (**WordDocument**, **PdfDocument**, **ExcelDocument**), making the system flexible and easy to extend with new document types without modifying existing code.